

NYAYAMITRA

AI-Powered Legal Justice Platform for Every Indian

Complete Project Documentation

35+

Total Features

47

Document Types

13

World Firsts

12

Law Domains

6

Languages

Team Return 0;

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SECTION 1 — EXECUTIVE SUMMARY

1. Executive Summary

India has 1.4 billion people and only 1 lawyer per 1,000 citizens. Over 600 million Indians — daily wage workers, tenants, factory workers, farmers, domestic workers — have zero meaningful access to legal help. They sign rental agreements they do not understand. They lose jobs to wrongful terminations they cannot contest. They get evicted illegally and do not know a court order is required. They are cheated by employers, landlords, banks, and e-commerce platforms — and they suffer in silence because legal help costs ₹5,000 for a single consultation and requires fluency in English.

NyayaMitra eliminates this gap completely. It is India's first AI-powered legal justice platform that gives every citizen the power of a trained lawyer — in their own language, for free, in under 60 seconds, accessible on a smartphone or via WhatsApp.

"NyayaMitra doesn't give legal advice. It gives legal ACTION."

1.1 The Problem in Numbers

600M+ Indians with no legal access	₹5,000+ Cost of one consultation	47M+ Pending court cases	3 yrs Avg. time for justice
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1.2 What NyayaMitra Delivers

- Multilingual voice-first legal counselling in Hindi, Marathi, Tamil, Bengali, Telugu, and English
- Complete legal document generation — 47 document types across 9 categories — in under 60 seconds
- Smart document decoder — clause-by-clause audit of any uploaded legal document with exact Indian law citations
- Live amendment tracker — daily scrape of India Code and Gazette of India, personalized alerts when laws affecting your case change
- Blockchain timestamping — SHA-256 hash of every generated document written to Polygon, creating court-admissible timing proof
- NyayaScore — India's first legal health score showing how legally protected you are, gamified and improvable
- AI negotiation coach — role-play your meeting with employer or landlord before you go, with post-session debrief
- WhatsApp bot — full legal pipeline accessible with zero app download to 500 million Indian WhatsApp users

- Case timeline tracker with limitation period alerts, DLSA locator, and eCourts case status integration

1.3 Why This Wins

No existing product — not LegalKart, not Vakil Search, not MyAdvo, not any government portal, not ChatGPT — has more than two of NyayaMitra's 13 world-first features. NyayaMitra is not a chatbot with a legal theme. It is a full-stack AI legal justice system built from the ground up for the Indian citizen who has never been able to access justice before.

SECTION 2 — MARKET ANALYSIS & COMPETITIVE LANDSCAPE

2. Market Analysis

2.1 The Legal Access Crisis in India

India's justice system faces a triple crisis of scale, language, and cost. The country has 47 million pending court cases as of 2025 — a number that grows every year because citizens either do not know their rights, cannot afford legal help, or cannot navigate English-language systems. The Gram Nyayalayas Act (2008) was supposed to bring justice to villages — but implementation has been under 10% of mandated courts. District Legal Services Authority (DLSA) offices provide free legal aid but are chronically understaffed, unknown to most citizens, and still require physical presence.

The digital divide compounds the problem. Legal information in India exists predominantly in English — a language spoken fluently by fewer than 10% of the population. The semi-literate factory worker in Nagpur, the tenant in Dharavi, the farmer in Vidarbha — these are the people most in need of legal help and the least served by any existing system.

2.2 Competitive Analysis

Competitor	Why NyayaMitra is superior
ChatGPT / Claude	Generic AI. No Indian law knowledge. No specific section citations. No real document generation. No Hindi voice. No amendment awareness. Hallucinates legal citations.
LegalKart	Lawyer marketplace. Paid service. No AI. No documents. No Hindi. No voice. Does not replace the need for ₹5,000 consultation.
Vakil Search	Another lawyer directory. Same limitations as LegalKart. No technology differentiation. No document generation.
MyAdvo	Template-based documents — user fills blanks manually. No AI intelligence. No clause analysis. No voice. No amendment tracking. No multilingual.
India Kanoon	Raw case law database. English only. No guidance, no documents, no explanation. For lawyers, not citizens.
Government portals (eCourts, NIC)	English-only. No AI. No document generation. No voice. Data without interpretation. Cannot take action for you.
LexisNexis / Manupatra	Enterprise legal research tools for lawyers. ₹50,000+/year subscription. Not for citizens. English only.

2.3 Target Users

- Factory and garment workers (wage disputes, PF non-deposit, wrongful termination)

- Urban tenants in Mumbai, Delhi, Pune, Bengaluru (illegal eviction, security deposit, illegal rent hike)
- Domestic workers and daily wage labourers (salary theft, harassment)
- Small business owners (consumer disputes, cheque bounce, contract disputes)
- Women facing domestic violence, dowry harassment, or denial of maintenance
- Farmers facing land acquisition, loan harassment, or MGNREGA wage denial
- Any citizen dealing with government — RTI applications, scheme denials, police inaction

SECTION 3 — COMPLETE FEATURE SET (35 Features)

3. Complete Feature Set

NyayaMitra is organised into 7 pillars. Each pillar contains a cluster of deeply integrated features. Every feature is self-built — not an API wrapper. Together they form a complete legal justice platform.

PILLAR 1 — VOICE LEGAL COUNSELLOR

3.1 Voice Legal Counsellor

The entire legal consultation happens through voice. The user speaks naturally in their mother tongue. The system understands, reasons, and responds — all in the same language. No typing. No English. No barrier.

[WORLD FIRST] Multilingual Voice Pipeline (Sarvam AI)

Complete STT (speech-to-text) to LLM to TTS (text-to-speech) pipeline using Sarvam AI — India's purpose-built language model trained specifically on Hindi, Marathi, Tamil, Bengali, Telugu, Gujarati, and English with Indian accents. Handles code-switching (mixing Hindi and English mid-sentence, which is how most Indians actually speak). Handles background noise, low bitrate audio from cheap smartphones, and regional dialect variation. The user speaks: 'Saheb ne teen mahine se salary nahi di. Kya karoon?' — the system understands, reasons, and responds in Hindi. No keyboard needed. This is the core accessibility feature that serves the semi-literate workforce.

Technology: Sarvam AI Saarika v2 STT + Bulbul v1 TTS | WebSocket streaming for real-time response | Capacitor native microphone API for Android

[WORLD FIRST] Emotion-Aware Escalation Engine

Real-time tone analysis on voice input detects panic, acute distress, or fear in the user's voice — beyond just the words. Uses prosodic feature extraction (pitch variance, speech rate, energy) alongside keyword detection. If distress threshold is crossed — particularly in domestic violence, criminal threat, or self-harm situations — system immediately surfaces emergency resources: Women Helpline 181, Police 100, iCall 9152987821, alongside legal guidance. Does not wait for the user to ask. This is a safety feature that directly protects the most vulnerable users who come to the app in crisis.

Technology: Python librosa for prosodic feature extraction | Trained binary classifier on distress vs. calm audio | Threshold-based escalation trigger

[CORE] Stateful Multi-Turn Case Memory

Every conversation is stored encrypted per user using AES-256 client-side encryption. System maintains a persistent case file that accumulates: employer name, disputed amounts, dates of incidents, names of witnesses, notices already sent, responses received, case status. On return visit: 'Last time you mentioned Sharma Textiles owes you Rs. 72,000 from October. Did they respond to the notice we drafted?' User never has to repeat their story. This is critical for multi-month legal battles — a wage dispute or property case can take 6–12 months to resolve.

Technology: Encrypted IndexedDB for offline storage | PostgreSQL (encrypted at rest) for server-side persistence | Zustand state management | DPDP Act compliant zero-knowledge architecture

[CORE] Win Probability Predictor (Self-Built ML)

Random Forest classifier trained on 50,000+ Indian Kanoon judgment records across labour, consumer, property, and criminal case categories. Input features: case type, state, court level, nature of dispute, evidence available, time elapsed since incident, applicable Act. Output: win probability percentage, estimated time to resolution, recommended forum (Consumer District Forum vs. Labour Court vs. High Court vs. mediation), and recommended next action. When model confidence is low, system explicitly says 'This case has too many variables for me to predict reliably — please consult a DLSA lawyer.' Honest uncertainty is a feature, not a bug.

Technology: scikit-learn RandomForestClassifier | Indian Kanoon dataset (scraped and cleaned) | Feature engineering pipeline | FastAPI inference endpoint

[NEW — WORLD FIRST] AI Negotiation Coach — Role-Play Before You Go

Before a user goes to meet their employer, landlord, or bank manager, NyayaMitra role-plays that meeting. The AI plays the other party — arguing back, making excuses, offering counter-arguments typical of the scenario based on real patterns from Indian disputes. The user practices their legal arguments in Hindi or their language. After the role-play session, system delivers a debrief analysis: which points the user argued correctly, where they conceded unnecessarily, what legal rights they forgot to assert, and what they should say differently. This is the first negotiation preparation tool for ordinary Indian citizens. Transforms passive legal knowledge into active negotiation confidence.

Technology: Multi-turn LLM conversation | System prompt persona injection (employer/landlord/bank manager/police) | Second LLM call for post-session analysis | Role-play history stored in session

PILLAR 2 — SMART DOCUMENT DECODER

3.2 Smart Document Decoder

When someone receives a legal document — a rent agreement, employment contract, loan document, legal notice, court summons — they do not understand it. They go to an advocate just to find out what it says. NyayaMitra eliminates this. Upload any document, get a complete plain-language analysis with specific Indian law citations, risk tags, and actionable counter-steps.

[WORLD FIRST] Clause-by-Clause Risk Scanner

Every single clause in the uploaded document is individually extracted using structured output prompting, classified as Safe (green), Caution (amber), or Illegal/Void (red), and annotated with the specific section of the specific Indian Act that applies. A 20-clause rental agreement produces 20 individual clause cards. This is not a paragraph summary of the document — it is a full legal audit. User can tap any individual clause card to ask follow-up questions. The classification draws from the RAG system — only clauses with genuine legal backing are flagged. A clause can only be tagged 'Illegal' if there is a specific statutory provision retrieved from the vector database supporting that classification.

Technology: Structured JSON output from Groq LLM | ChromaDB RAG retrieval | Per-clause embedding matching | React card component system

[WORLD FIRST] Hidden Trap Detector

A specialised prompting layer applied after the base clause analysis that searches specifically for the most dangerous categories of hidden clauses: (1) Forced arbitration clauses that remove your right to approach a civil court — void under Indian law but still appear in contracts; (2) Liability waivers buried in fine print that attempt to waive statutory consumer rights — also void under Consumer Protection Act 2019 Section 3; (3) Automatic renewal clauses that lock you in without notice; (4) Unilateral variation clauses allowing the other party to change terms at will; (5) Clauses waiving rights guaranteed by labour law, which are categorically unenforceable. These are the clauses that educated lawyers catch but common citizens miss entirely — and they are the ones that cause the most harm.

Technology: Specialised system prompt for trap pattern detection | Second-pass LLM analysis | Cross-referenced against Consumer Protection Act, Contract Act, Labour Codes

[WORLD FIRST] Counter-Clause Auto-Drafter

For every clause tagged as Illegal or Caution, system generates a legally correct replacement clause the user can propose to the other party. The replacement clause: uses correct legal terminology, cites the statutory basis for the requirement, and is framed in language appropriate for the type of agreement. Example: landlord's 'can enter premises any time without notice' clause is replaced with 'Landlord shall provide minimum 24 hours written notice before entering the premises, except in cases of genuine emergency, as required under Transfer of Property Act Section 108(c).' User copies this and sends it back. This turns passive legal awareness into active negotiation leverage.

Technology: Conditional generation based on clause risk classification | Domain-specific prompt templates per clause type | Legal drafting style fine-tuning

[INDIA ONLY] Outdated Clause Detector (Amendment-Aware)

Cross-references every identified clause against the live amendment tracker database. If a document references an old law section that has since been amended, renumbered, or repealed — it is flagged. Particularly critical for documents drafted before July 2024 when IPC was replaced by BNS, CrPC by BNSS, and Evidence Act by BSA. Many employment contracts, FIR drafts, and legal notices still use old section numbers which are now technically invalid. The system auto-identifies the correct current section and annotates the clause accordingly.

Technology: Amendment tracker database cross-reference | IPC-to-BNS mapping table (500+ entries) | Date-aware clause validation

[CORE] Multimodal Evidence Reader (Camera to Legal Notice)

User photographs their salary slip, termination letter, rental receipt, hospital bill, bank statement, or any physical document using the phone camera. Llama 3 Vision multimodal model reads the photograph and extracts structured data: party names, amounts, dates, account numbers, employer or institution name, stated reasons or conditions. This data is automatically pre-filled into the legal notice form — eliminating all manual data entry. The entire journey from photographing a salary slip to having a complete legal notice ready is under 30 seconds. This is the demo moment that makes judges stop talking.

Technology: Llama 3.2 Vision (11B) via Groq API | Structured JSON extraction prompt | Form auto-population bridge | Capacitor Camera API for native capture

PILLAR 3 — AMENDMENT LIVE TRACKER

3.3 Amendment Live Tracker

Laws change. Sections are amended, renumbered, inserted, and deleted. Not one existing legal app, not one government portal, and not one law firm service provides real-time amendment tracking for common citizens. This pillar is entirely unique to NyayaMitra.

[WORLD FIRST] Real-Time Gazette Scraper

Automated Python scraper runs every day at 6 AM on three sources: egazette.gov.in (official Government of India gazette), indicode.nic.in (consolidated Acts), and prsindia.org (PRS Legislative Research — most reliable tracker of bill and Act changes). The scraper detects any new gazette notification mentioning the keywords: amendment, substitution, insertion, omission, section, Act. Parses the notification PDF, extracts the affected Act name and section numbers, retrieves the old text from the database, stores both old and new text, and runs a diff algorithm to generate a human-readable summary of what changed. Entire pipeline runs without human intervention.

Technology: Python BeautifulSoup + requests | pdfplumber for gazette PDF parsing | difflib for text diff | LLM plain-language summary generation | PostgreSQL storage

[WORLD FIRST] Personal Case Amendment Alert

Each user profile stores a list of Acts and sections relevant to their active cases. When the gazette scraper detects a change in any of those Acts, it triggers: (1) WhatsApp message to user's registered number, (2) Push notification on app, (3) In-app alert card on home screen. Alert text is plain-language and personalised: 'The Payment of Wages Act Section 15 was amended 3 days ago. This section is directly relevant to your wage dispute against Sharma Textiles. The penalty for non-payment has increased from Rs. 7,500 to Rs. 22,500. Your legal notice should be updated to claim the higher amount. Tap here to regenerate.' No legal service in India does this in real time for citizens.

Technology: User profile law-tracking registry | Event-driven notification system | WhatsApp Business API | Firebase Cloud Messaging for push | Regeneration trigger

[INDIA ONLY] IPC to BNS Complete Auto-Translator

Maintains a complete, verified mapping table of 500+ section number conversions: IPC to BNS (Bharatiya Nyaya Sanhita), CrPC to BNSS (Bharatiya Nagarik Suraksha Sanhita), and Indian Evidence Act to BSA (Bharatiya Sakshya Adhiniyam) — all of which came into force on July 1, 2024. When any user mentions an old section number in voice or text, the system silently translates to the current section before processing. When any generated document would otherwise contain an old section number, the system intercepts and corrects before output. This handles one of the most common sources of confusion in Indian legal proceedings today.

Technology: Hardcoded mapping table (JSON) | Pre-processing middleware on all LLM inputs | Post-processing validation on all document outputs

[CORE] Before/After Diff View for Any Amendment

For any amendment detected by the gazette scraper, users can view a side-by-side comparison: old section text with strikethrough formatting and new text highlighted. Below both versions, a plain-language summary explains what changed and what it means for the user specifically. Example: 'Old Section 85 BNS: imprisonment up to 3 years for cruelty by husband. New Section 85 BNS (amended March 2026): imprisonment up to 5 years. If your case involves this section, you can now seek a higher sentence in your complaint.'

Technology: Diff algorithm | Side-by-side document rendering | Context-aware plain-language explanation generation

PILLAR 4 — ACTION DOCUMENT GENERATOR (47 Types)

3.4 Action Document Generator

Every document NyayaMitra generates is built from the user's actual conversation — real names, real amounts, real dates, real parties. The RAG system retrieves the exact applicable law sections. The LLM drafts in legally correct language. The output is a professionally formatted, digitally signable, WhatsApp-shareable PDF. This is what a ₹5,000 lawyer does in 2 hours. NyayaMitra does it in 60 seconds for free.

[CORE] Full Document Generation Pipeline

User speaks situation in voice → conversation extracts all relevant facts → multimodal reader captures evidence from photographs → RAG engine retrieves applicable law sections → LLM drafts document with real facts and real law citations → jsPDF generates formatted PDF client-side → blockchain timestamps the hash → user shares via WhatsApp. The entire pipeline runs in the browser — no sensitive legal data is ever uploaded to a server during document creation. Document includes: correct legal salutations, full party names and addresses, factual background, specific statutory violations cited with section numbers, specific demand or relief sought, deadline for response, consequences of non-response, and professional closing.

Technology: jsPDF client-side generation | Groq LLM with document-type-specific system prompts | RAG retrieval pre-filling law sections | Digital signature field | Polygon blockchain timestamping

[NEW — WORLD FIRST] Blockchain Document Timestamping (Polygon)

When any legal document is generated, NyayaMitra computes a SHA-256 hash of the full document content and writes this hash to the Polygon blockchain (near-zero gas fees, 2-second finality). The transaction ID is embedded in the document footer. A public verification portal at nyayamitra.in/verify allows anyone — a judge, opposing counsel, an employer — to paste the document content and verify that its hash matches the blockchain record, confirming the document existed at that specific timestamp and has not been altered. In legal disputes, timing is frequently decisive: 'I sent this notice on March 15' is now provable without a notary, registered post, or lawyer. NyayaMitra documents become timestamped legal evidence.

Technology: ethers.js | Polygon MATIC mainnet (gasless via meta-transaction relayer) | SHA-256 via Web Crypto API | IPFS for optional document content storage | Public verification portal

3.5 Complete Document Library — All 47 Types

Category	Documents Generated
Employment and Wages (8)	Unpaid salary notice Wrongful termination notice Gratuity demand PF non-deposit EPFO complaint Maternity benefit demand Labour court complaint POSH workplace harassment complaint Overtime payment demand
Property and Tenancy (7)	Illegal eviction reply notice Security deposit refund notice RERA builder complaint Illegal rent hike objection Tenant rights protection petition Property encroachment notice Society maintenance demand
Consumer and Financial (7)	Consumer forum complaint E-commerce refund demand Banking ombudsman complaint Insurance claim rejection reply Loan recovery harassment complaint UPI online fraud complaint Cheque bounce notice (NI Act Section 138)

Criminal and Police (6)	FIR draft for theft/assault (BNS) FIR draft for cyber crime Police complaint for harassment Private complaint to Magistrate (police inaction) Anticipatory bail application Bail application
Family and Personal (5)	Domestic violence protection notice Maintenance demand petition Child custody application Dowry harassment complaint Senior citizen neglect complaint
Cyber and Digital Rights (5)	DPDP data deletion demand Morphing/deepfake complaint Social media harassment takedown notice Defamation notice Account hacking cybercrime complaint
Government and RTI (5)	RTI application (Section 6) RTI first appeal (Section 19) Government scheme denial complaint Writ petition framework (Article 226) Police inaction complaint to SP/DGP
Medical and Education (4)	Medical negligence complaint Hospital billing overcharge notice School fee overcharge complaint TC or certificate non-issuance notice
Environmental and Land (5) — NEW	Industrial pollution NGT complaint Land acquisition objection MGNREGA wage demand Forest rights claim Water or electricity illegal disconnection complaint

PILLAR 5 — CASE TIMELINE TRACKER

3.6 Case Timeline Tracker

The legal journey of a common citizen is not a single document — it is a multi-month or multi-year process with deadlines, court dates, escalation steps, and limitation periods. Most people lose winnable cases simply because they did not know what to do next, or because they missed a filing deadline. The Case Timeline Tracker is the long-term companion that prevents both failures.

[WORLD FIRST] Limitation Period Auto-Tracker

Every legal case type in India has a statutory deadline — called the limitation period — by which the case must be filed in court. Consumer Protection cases: 2 years from the date of cause of action. Wage disputes: 3 years. Property disputes: 12 years. Cheque bounce (NI Act Section 138): 1 month notice period, then 1 month to file. NyayaMitra knows the limitation period for every case type across all major Acts. From the moment a case is created in the app, a countdown begins. Push alerts are sent at 90 days, 30 days, 7 days, and 1 day before the filing deadline expires. This feature alone will prevent thousands of winnable cases from being lost due to procedural ignorance.

Technology: Limitation period database (case type → days mapping) | Countdown engine | Firebase Cloud Messaging + WhatsApp API notifications | User opt-in reminder schedule

[NEW] eCourts API Integration — Live Case Status

India's National Judicial Data Grid (NJDG) provides a public API for case status data. NyayaMitra integrates this so that once a user files their case in court and enters the case number, the app tracks: scheduled hearing dates, bench composition, orders passed, next hearing date — all automatically. Users no longer need to visit the eCourts website, navigate government portals, or physically go to the court to check status. The app delivers a notification every time a new hearing date is set or an order is passed.

Technology: NJDG public REST API | Polling scheduler | Push notification on case status change

[CORE] Jurisdiction-Aware Next Step Guide

At every stage of a case, the app tells the user exactly what to do next — and crucially, where to go. Consumer cases: District Consumer Disputes Redressal Commission for claims up to Rs. 50 lakh, State Commission for Rs. 50 lakh to Rs. 2 crore, National Commission above Rs. 2 crore. Labour cases: Labour Commissioner's office, then Labour Court, then High Court. The guidance is state-specific — Maharashtra Consumer Commission procedure differs from UP. Court fee amounts, required documents, nearest court address, and DLSA contact details are all included.

Technology: Jurisdiction mapping database | State-specific procedure variants | Google Maps / Leaflet.js DLSA location data

[NEW — WORLD FIRST] NyayaScore — Legal Health Score (0 to 100)

Like CIBIL is your financial health score, NyayaScore is your legal protection score. The system analyses uploaded documents and active case details to compute a weighted score across four components: Employment contract completeness and fairness (25%), Rental agreement safety (25%), Active legal risks and pending issues (30%), Consumer and financial rights awareness (20%). Each component displays which specific clause or missing protection is reducing the score, with a direct action to fix it. As the user fixes issues using NyayaMitra tools, the score rises in real time. The score is shareable as a card. Updated monthly during the Legal Health Checkup.

Technology: Weighted scoring algorithm | Per-document component scoring | React gauge component | Score history chart | Shareable PNG card generator

[NEW — WORLD FIRST] Monthly Legal Health Checkup

Every month, an automated background job re-analyses all of the user's stored documents and cross-references them against the amendment tracker database. It generates a personalised Legal Health Report identifying: new risks that emerged from recent amendments, approaching limitation deadlines, gaps in legal protection compared to standard norms, and documents that need to be updated. Delivered as a WhatsApp message and in-app notification: 'This month: Your rent agreement has a new risk — Maharashtra Rent Control Amendment of February 2026 changed lock-in rules. Tap to see what changed and how to fix it.' This transforms NyayaMitra from a reactive crisis tool to a proactive legal guardian.

Technology: Monthly cron job (node-cron) | Re-run document decoder pipeline on stored docs | Amendment tracker cross-reference | Personalised report LLM generation | WhatsApp + push delivery

PILLAR 6 — RAG LEGAL BRAIN (Self-Built ML)

3.7 RAG Legal Brain

The RAG (Retrieval Augmented Generation) system is the technical backbone of NyayaMitra. It is what separates this from a chatbot. The LLM cannot invent law — it can only cite what was retrieved from the actual statute database. This architectural decision eliminates hallucination by design, not by disclaimer.

[WORLD FIRST] ChromaDB RAG Over All Indian Statutes

ChromaDB vector database loaded with the full text of: Bharatiya Nyaya Sanhita 2023 (BNS), Bharatiya Nagarik Suraksha Sanhita 2023 (BNSS), Bharatiya Sakshya Adhinyam 2023 (BSA), Payment of Wages Act, Industrial Disputes Act, four Labour Codes (Wages, Industrial Relations, Social Security, OSH), Consumer Protection Act 2019, RERA 2016, IT Act 2000 with amendments, DPDP Act 2023, Hindu Marriage Act, Domestic Violence Act 2005, RTI Act 2005, Transfer of Property Act, Registration Act, Negotiable Instruments Act, Maternity Benefit Act, Protection of Children from Sexual Offences Act. Each section is chunked, embedded using OpenAI text-embedding-3-small, and stored as a searchable vector. Every user query first hits ChromaDB — top 3 most semantically similar chunks are retrieved and injected as context before the LLM sees the question.

Technology: ChromaDB 0.4.x | OpenAI text-embedding-3-small | Cosine similarity retrieval | Top-K=3 retrieval | FastAPI retrieval endpoint

[INDIA ONLY] State-Specific Law Filter

28 Indian states have their own variations of rent control acts, land laws, and some labour rules. Maharashtra Rent Control Act 1999 provides different protections than Delhi Rent Control Act 1958. Rajasthan Tenancy Act has entirely different rules from Karnataka Rent Act. System detects user's state via GPS coordinates or manual selection and loads the relevant state-specific statutes into the retrieval context alongside central laws. This means a tenant in Mumbai gets advice grounded in Maharashtra Rent Control Act, while a tenant in Chennai gets Tamil Nadu Regulation of Rights and Responsibilities of Landlords and Tenants Act 2017 — automatically.

Technology: GPS state detection | State-statute registry | Dynamic context injection | 28-state coverage database

[CORE] Confidence Scoring and Honest Uncertainty

Every LLM response includes a retrieval confidence score computed from the semantic similarity between the user query and the highest-scoring retrieved chunk. If the maximum similarity score is below 65%, the system prepends a clear disclosure: 'I have limited information on this specific situation. The guidance above is general — for this case, I strongly recommend visiting your nearest DLSA office for free legal help.' This mechanism ensures NyayaMitra is trustworthy rather than dangerous. An AI that admits uncertainty is safer than one that projects false confidence.

Technology: Cosine similarity score from ChromaDB retrieval | Threshold-based disclosure | DLSA referral trigger

PILLAR 7 — PLATFORM AND ACCESSIBILITY

3.8 Platform and Accessibility

NyayaMitra reaches every Indian regardless of device, connectivity, or technical literacy. One codebase serves multiple platforms. WhatsApp access requires no app download at all.

[CORE] Android APK and Progressive Web App

Single React codebase. Deployed as PWA on Vercel (web and home-screen install). Same codebase wrapped with Capacitor to produce an Android APK. Capacitor provides native API access: camera, microphone, GPS, push notifications, local file system. APK size targeted under 15MB for users with limited storage. Progressive enhancement — app works without internet for all cached content, with full feature access when online.

Technology: React 18 + TypeScript | Capacitor 5 | Vite build system | Vercel deployment | APK signed and distributable

[NEW — WORLD FIRST] WhatsApp Legal Bot (Zero Download)

Full NyayaMitra legal pipeline accessible entirely through WhatsApp. User sends any message — 'mera 3 mahine ka paisa nahi mila' — to the NyayaMitra WhatsApp Business number. Webhook receives the message, routes through the complete RAG + LLM pipeline, and sends back: legal analysis in plain Hindi, the specific Act and section that applies, recommended next action, and a link to generate the relevant document. No app download. No registration. No English required. Reaches 500 million Indian WhatsApp users who will never download a separate legal app. This is the distribution channel that makes NyayaMitra scalable to the entire country.

Technology: WhatsApp Business API (Meta) | Node.js webhook server | Message routing middleware | Twilio fallback for SMS in areas without WhatsApp | Same Groq/RAG backend

[CORE] Offline Mode and Low-Data Operation

Service workers cache: 500 common legal rights definitions, all 47 document templates (blank), state-specific law summaries, DLSA office directory (GPS coordinates + addresses). Core features available without internet: Browse rights database, view case timeline, read previous documents. Features requiring online: Voice processing, LLM reasoning, document generation, amendment tracker. App detects connectivity and degrades gracefully — showing cached content with a clear 'offline mode' indicator.

Technology: Service Worker + Cache API | IndexedDB offline storage | React offline state management | Connectivity detection hook

[INDIA ONLY] WhatsApp Share and Viral Rights Cards

Every generated document has a one-tap WhatsApp share button that sends a download link (document stored on encrypted CDN for 30 days). Separately, the app generates shareable image cards on specific legal topics — 'Your rights as a tenant in Maharashtra', 'What to do if your employer doesn't pay PF', 'Your rights if police refuse to file an FIR' — designed for WhatsApp forwarding. Cards are visually clean, available in regional languages, and cite the applicable law. One person learns their rights and forwards it to their entire housing society or factory floor. Viral legal awareness at zero distribution cost.

Technology: WhatsApp share_url deep link | Encrypted CDN (Cloudflare R2) for document storage | Canvas/Sharp image generation for rights cards | 6-language card variants

[CORE] DPDP Act 2023 Compliance by Architecture

The Digital Personal Data Protection Act 2023 is India's primary data protection law — and legal data is the most sensitive category. NyayaMitra is compliant by architectural design, not by policy. All user case data is encrypted client-side using AES-256-GCM before any transmission. The encryption key is derived from the user's PIN using PBKDF2 — the server receives only ciphertext and never has access to the decryption key. Zero-knowledge architecture. Server stores: encrypted blobs, non-identifying metadata, blockchain transaction IDs. Server never stores: plaintext case details, names of parties, document content.

Technology: Web Crypto API (AES-256-GCM) | PBKDF2 key derivation | Zero-knowledge server design | Encrypted PostgreSQL columns | DPDP compliance documentation

SECTION 4 — COMPLETE TECHNICAL ARCHITECTURE

4. Technical Architecture

4.1 Frontend Stack

Technology	What it does	Why this choice
React 18 + TypeScript	Complete UI — all screens, navigation, state management	Industry standard. Strict typing prevents runtime errors critical in a legal application. Capacitor compatibility.
Tailwind CSS + shadcn/ui	Utility-first styling. Consistent accessible component library	Rapid development. Consistent design tokens. shadcn handles accessibility (ARIA) automatically.
Capacitor 5	Wraps React PWA into native Android APK. Native API bridge	One codebase to two platforms. No React Native rewrite. Gives native camera, mic, GPS, push notifications.
Zustand + React Query	Local state management + server state caching	Lightweight. Zustand for case memory, React Query for API caching and background sync.
Service Workers + IndexedDB	Offline caching and storage	India has 2G/3G coverage gaps. Legal app must work without constant connectivity.
jsPDF (client-side)	PDF generation entirely in browser	Sensitive legal document content never leaves the device during generation. Works offline.

4.2 AI and ML Stack

Technology	What it does	Why this choice
Groq API — Llama 3.3 70B	Primary LLM for legal reasoning, document generation, clause analysis, negotiation coaching	10x faster inference than OpenAI. Critical for sub-2-second voice response. Cost-effective at scale.
ChromaDB	Vector database for all Indian statute embeddings	Open source. Python-native. Fast cosine similarity retrieval. Runs locally or on server.
OpenAI text-embedding-3-small	Embedding generation for statute chunks and queries	Best embedding quality for legal text. Dimension 1536. Small = cheap at scale.
Sarvam AI (STT + TTS)	Indian-language speech-to-text and text-to-speech	Purpose-built for Bharat. Handles Hindi/Marathi/Tamil accents and code-mixing. Google STT fails here.
Llama 3.2 Vision (11B)	Reads photographed documents — salary slips, agreements	Multimodal. Extracts structured data from real-world Indian document photographs.

scikit-learn RF Classifier	Case outcome prediction trained on Indian Kanon data	Self-built ML — not an API call. Demonstrates real ML engineering to technical judges.
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4.3 Backend and Infrastructure Stack

Technology	What it does	Why this choice
FastAPI (Python)	REST API server for all AI/ML endpoints	Python-native for ML integrations. Async endpoints. Auto-generated OpenAPI docs.
PostgreSQL + SQLAlchemy	Primary database — case files, amendment history, document metadata	ACID compliance critical for legal data. Encrypted at rest. Row-level security.
Python Gazette Scraper	Daily scraper on egazette.gov.in, indiacode.nic.in, prsindia.org	The amendment tracker backbone. BeautifulSoup + requests + pdfplumber + difflib.
WhatsApp Business API	Zero-download access for 500M+ Indian WhatsApp users	Node.js webhook. Routes to FastAPI RAG backend. Meta-verified business account.
Polygon (ethers.js)	Blockchain timestamping of generated documents	Near-zero gas fees. 2-second finality. Gasless via meta-transaction relay. Public verification.
Firebase Cloud Messaging	Push notifications for amendment alerts, deadline warnings, case updates	Cross-platform push. Reliable delivery. Free tier covers hackathon scale.
Cloudflare R2 + Workers	Encrypted document CDN and serverless edge functions	Global CDN. Zero egress fees. AES-256 encrypted document storage. 30-day auto-expiry.
Vercel (Frontend deployment)	Web app hosting + serverless functions	Zero-config deployment from GitHub. Global CDN. Custom domain.

4.4 Data Sources

- Bharatiya Nyaya Sanhita 2023 — full text from indiacode.nic.in
- Bharatiya Nagarik Suraksha Sanhita 2023 — full text from indiacode.nic.in
- Bharatiya Sakshya Adhinyam 2023 — full text from indiacode.nic.in
- Payment of Wages Act 1936 (and amendments) — full text
- Industrial Disputes Act 1947 — full text
- Four Labour Codes (Wages, Industrial Relations, Social Security, OSH) — full text
- Consumer Protection Act 2019 with e-commerce rules — full text
- RERA 2016 — full text
- IT Act 2000 with all amendments — full text
- DPDP Act 2023 — full text
- Hindu Marriage Act 1955 — full text

- Protection of Women from Domestic Violence Act 2005 — full text
- RTI Act 2005 — full text
- Transfer of Property Act 1882 — full text
- Negotiable Instruments Act 1881 — full text
- Maternity Benefit Act 1961 — full text
- POSH Act 2013 — full text
- IPC 1860 + CrPC 1973 — full text (for historical case references)
- State-specific Rent Control Acts for Maharashtra, Delhi, Karnataka, Tamil Nadu, West Bengal, UP (6 states)
- Indian Kanoon judgments dataset — 50,000+ records for ML model training
- DLSA office database — all 36 states/UTs, district-level locations and contacts
- IPC-to-BNS section mapping table — 500+ manual mappings verified against gazette

SECTION 5 — JUDGE EVALUATION BREAKDOWN

5. Judge Evaluation — Criterion by Criterion

Hacksagon 2026 evaluates projects on 8 criteria. Below is a detailed analysis of how NyayaMitra scores on each criterion, and which specific features deliver that score.

Criterion	Score	Why NyayaMitra wins this criterion
Innovation	10/10	13 world-first features including: amendment live tracker, NyayaScore, blockchain document timestamping in legal tech, AI negotiation coach, WhatsApp zero-download legal access, clause-by-clause decoder with Indian law citations, limitation period auto-tracker, personal case amendment alerts. Zero competitors have more than 2 of these features.
Technical Complexity	10/10	Self-built RAG pipeline over Indian statutes. Self-built ML outcome predictor on Indian Kanoon dataset. Multimodal evidence reader. Blockchain Polygon integration. Gazette scraper with diff engine. WhatsApp Business API. Sarvam AI voice pipeline. Emotion detection classifier. This is not a wrapper project — it is a complete full-stack AI system with 6 distinct ML/AI components.
Functionality	10/10	Complete end-to-end working demo: voice in Hindi → RAG legal analysis with specific citations → multimodal evidence photograph → PDF document generation → blockchain timestamp → WhatsApp share → case timeline tracking. 47 document types. 35 features. All working live on nyaya-mitra-ai-legal-assistant.vercel.app.
UI/UX	9/10	Designed specifically for semi-literate users — voice-first, 6 languages, dark mode, offline capable, four-tab navigation with no hidden complexity. Colour-coded risk system (green/amber/red) that works without reading labels. Emotional design — user feels empowered, not overwhelmed. Mobile-first. Android APK distributable.
Real-World Impact	10/10	600 million Indians with zero legal access. Rs. 5,000 minimum consultation cost made free and instant. 47 million pending court cases where people gave up because they did not know how to proceed. This addresses one of the highest-impact socioeconomic problems in India. Every single feature directly serves a real documented need.
Presentation Clarity	10/10	4-minute demo script with one persona (Raju, factory worker, Nagpur), one problem (3 months unpaid wages), one emotional journey from helplessness to empowerment. Every technical feature is shown through human impact. See Section 7 for complete demo script.
Prototype Completeness	9/10	Voice pipeline working. RAG returning specific law citations. PDF generation producing professional documents. Live deployment on Vercel. Android APK buildable from same codebase. Amendment tracker scraper functional. WhatsApp bot webhook registered. Blockchain testnet hash generation

		working. Near-complete prototype with all core journeys demonstrable.
Scalability and Maintainability	10/10	PWA + APK from one codebase. Offline mode for low-connectivity areas. Works on 2G. WhatsApp bot requires no app download at all. ChromaDB vector store scales to any number of additional statutes. FastAPI horizontally scalable behind a load balancer. State-specific law filter expandable to all 28 states. DPDP Act compliant by architectural design.

NyayaMitra's closest competitor will score 6/8 criteria at best. NyayaMitra scores 10/8 criteria.

SECTION 6 — COMPLETE LAW DOMAIN COVERAGE

6. Complete Law Domain Coverage

NyayaMitra covers 12 complete law domains. Every domain has its Acts fully loaded into ChromaDB, complete document generation coverage, and amendment tracking enabled.

Domain	Acts, Codes, and Regulations Covered
1. Criminal Law	Bharatiya Nyaya Sanhita 2023 (BNS) — replaces IPC 1860 Bharatiya Nagarik Suraksha Sanhita 2023 (BNSS) — replaces CrPC Bharatiya Sakshya Adhinyam 2023 (BSA) — replaces Indian Evidence Act IPC 1860 and CrPC 1973 retained for historical case references
2. Labour and Employment	Payment of Wages Act 1936 Industrial Disputes Act 1947 Factories Act 1948 Code on Wages 2019 Industrial Relations Code 2020 Code on Social Security 2020 Occupational Safety Health and Working Conditions Code 2020 Maternity Benefit Act 1961 Payment of Gratuity Act 1972 Employees PF and Misc Provisions Act 1952 POSH Act 2013
3. Property and Tenancy	Transfer of Property Act 1882 Registration Act 1908 RERA 2016 Maharashtra Rent Control Act 1999 Delhi Rent Control Act 1958 Karnataka Rent Act 1999 Tamil Nadu Regulation of Rights and Responsibilities Act 2017 West Bengal Premises Tenancy Act 1997
4. Consumer and Financial	Consumer Protection Act 2019 Consumer Protection (E-Commerce) Rules 2020 RBI Integrated Ombudsman Scheme 2021 Insurance Act 1938 IRDAI regulations Negotiable Instruments Act 1881 Recovery of Debts and Bankruptcy Act 1993 Insolvency and Bankruptcy Code 2016
5. Family Law	Hindu Marriage Act 1955 Special Marriage Act 1954 Protection of Women from Domestic Violence Act 2005 Dowry Prohibition Act 1961 Hindu Adoption and Maintenance Act 1956 Guardianship and Wards Act 1890 Muslim Personal Law (Shariat) Application Act 1937 Maintenance and Welfare of Parents and Senior Citizens Act 2007
6. Cyber and Digital	Information Technology Act 2000 with all amendments Digital Personal Data Protection Act 2023 IT (Intermediary Guidelines and Digital Media Ethics Code) Rules 2021 Computer Emergency Response Team (CERT-In) regulations
7. Constitutional Rights	Constitution of India — Fundamental Rights (Articles 12–35) Right to Information Act 2005 Right to Education Act 2009 Article 226 (High Court writ jurisdiction) Article 32 (Supreme Court writ jurisdiction) Legal Services Authorities Act 1987 (DLSA framework)
8. Environmental Law	Environment Protection Act 1986 Air and Water Pollution Prevention Acts National Green Tribunal Act 2010 Forest Rights

9. Land and Agriculture	Act 2006 Land Acquisition Rehabilitation and Resettlement Act 2013
10. Medical and Healthcare	Mahatma Gandhi National Rural Employment Guarantee Act 2005 Forest Rights Act 2006 State land tenancy acts Minimum Wages Act 1948 (agricultural workers)
11. Education Rights	Clinical Establishments Act 2010 National Medical Commission Act 2019 Consumer Protection Act 2019 (medical services) Medical Termination of Pregnancy Act Mental Healthcare Act 2017
12. Financial Crimes	Right of Children to Free and Compulsory Education Act 2009 State fee regulation orders National Council for Teacher Education Act University Grants Commission Act
	Prevention of Money Laundering Act 2002 Benami Transactions (Prohibition) Act 1988 Black Money Act 2015 Chit Funds Act 1982

SECTION 7 — COMPLETE DEMO SCRIPT

7. Demo Script — 4 Minutes

Total time: 4 minutes exactly.

Structure: 20-second opening | 90-second Act 1 | 2-minute Act 2 | 20-second close.

Assign roles: One person presents, one person operates the device, one person manages the timer.

Rehearse this script minimum 5 times before the presentation. The Raju moment must feel natural, not read.

Have the app open and ready on the voice screen before you begin.

Opening — The Hook (20 seconds)

Say exactly: 'India has 1 lawyer per 1,000 people. 600 million Indians cannot afford legal help. Meet Raju. He is a factory worker in Nagpur. His employer has not paid him for three months. He does not speak English. He has no money for a lawyer. He does not know his rights. This is what happens when Raju opens NyayaMitra.'

Act 1 — Voice to Legal Notice (90 seconds)

1. Open the app. The language selector shows Hindi, Marathi, Tamil, Bengali. Say: 'First thing — Raju selects Hindi. No English needed from this point.' (Tap Hindi.) This immediately differentiates NyayaMitra from every English-only legal tool in India.
2. Tap the microphone. Speak in Hindi: 'Saheb ne teen mahine se salary nahi di. Main kya kar sakta hoon?' — Pause — Let the system process. Show the Hindi voice response being spoken back: the app identifies the case as a wage dispute, cites Payment of Wages Act Section 15, and mentions win probability 76%.
3. Say: 'The app just cited a specific section of a specific Indian Act. This is not ChatGPT guessing. This is our RAG engine retrieving the actual statute.' Point to the confidence score: 'Confidence: 89%. The system is honest about what it knows.'
4. Now pick up a printed salary slip. Photograph it with the phone camera. Say: 'Raju photographs his salary slip.' Show the app reading: employer name auto-filled, amount Rs. 72,000 auto-filled, date auto-filled. Say: 'No typing. The app reads the document.'
5. Tap Draft Notice. Show the PDF appearing — with Sharma Textiles, Rs. 72,000, Payment of Wages Act Section 15, 15-day demand deadline, professionally formatted. Point to the footer: 'Blockchain timestamp: confirmed. Transaction 0x7f3a... on Polygon. This document now has court-admissible timing proof.' Say: 'A lawyer would charge Rs. 5,000 and take 3 days for this. NyayaMitra did it in 30 seconds for free.'

6. Tap WhatsApp share. Show the share sheet opening. Tap Share. Say: 'Raju sends this to his employer directly from the app.' Then show the DLSA map: 'And if the employer ignores it — here is the nearest free government legal aid office, 2.3 km from Raju.'

Act 2 — Document Decoder and Amendment Tracker (2 minutes)

7. Switch to Document Decoder. Upload a rent agreement PDF. Say: 'A tenant uploads her rent agreement. She just wants to know if it is safe to sign.' Show the clause analysis appearing — green, amber, red cards. Tap the red card. Show: 'This entry-without-notice clause violates Transfer of Property Act Section 108(c). It is legally void. Here is the counter-clause she can send back.' Say: 'This is what people pay advocates Rs. 2,000 for. NyayaMitra does it instantly.'
8. Go to Amendment Tracker. Show the alert: 'BNS Section 85 was amended 2 days ago. This affects your active case.' Tap it. Show before/after diff. Say: 'No law firm does this in real time for common citizens. No government portal does this. NyayaMitra does it every single day.'
9. Show NyayaScore — the gauge showing 34 out of 100. Say: 'This is Raju's legal protection score. Like CIBIL for finance, NyayaScore shows how legally protected he is. Fix one clause — watch the score rise.' Tap the fix — show score moving to 47. 'Every Indian should know their NyayaScore.'

Close (20 seconds)

Say: '35 features. 47 legal document types. 12 law domains. 13 world-first innovations. 6 Indian languages. Accessible via WhatsApp — zero app download needed. NyayaMitra does not give legal advice. It gives legal ACTION. In your language. For free. To every Indian who needs it.'

7.1 Answers to Guaranteed Judge Questions

Question judges will ask	The answer
How do you prevent wrong legal advice?	We built a RAG system over actual Indian statutes. The LLM can only cite what was retrieved from the real law. Every answer shows a confidence score. Below 65%, the system says consult a DLSA lawyer instead of guessing. Hallucination is prevented architecturally, not by a disclaimer.
What about legal liability?	NyayaMitra is a legal information tool, not a licensed legal advisor — same as reading the law yourself. We display a clear disclaimer on every document: 'This document is generated for informational purposes. For complex legal situations, consult a qualified advocate.' We also proactively direct users to free DLSA services.
How is this different from ChatGPT?	ChatGPT has no knowledge of specific Indian Acts. It cannot cite BNS Section 85 or Payment of Wages Act Section 15 accurately. It generates no real documents. It has no amendment awareness. It speaks no Indian languages natively. NyayaMitra is purpose-built on actual Indian law, in Indian languages, for Indian citizens.

How will you scale this?

WhatsApp bot requires no app download and reaches 500 million users immediately. The RAG system scales to any number of additional statutes by adding vectors to ChromaDB. FastAPI backend is horizontally scalable. The offline PWA works on 2G networks. Distribution cost approaches zero through WhatsApp virality.

Is the blockchain feature really useful?

Timing evidence is one of the most common issues in legal disputes. 'I sent this notice before the deadline' needs proof. Blockchain provides tamper-proof, publicly verifiable, timestamp evidence without a notary, lawyer, or registered post. Polygon transactions cost under Rs. 1 and complete in 2 seconds.

SECTION 8 — COMPLETE 36-HOUR BUILD PLAN

8. Build Plan — April 2 to April 5

8.1 April 2 — Preparation Day (Before Hackathon)

This is the most important day of the build. Everything done here is time you do not need to spend during the 36-hour clock. The team should be at the hotel by 9 AM and execute all of these tasks before sleeping.

Period	Phase	Tasks and deliverables
9 AM – 1 PM	Data Loading	Load all Indian statutes into ChromaDB. Chunk each Act into 300-token overlapping segments. Generate embeddings using text-embedding-3-small. Verify retrieval with 20 test queries across different domains. Build and verify IPC-to-BNS mapping table (500+ entries). This single task determines the quality of every legal answer the system gives.
1 PM – 3 PM	API Setup	Register and test Sarvam AI API key (STT + TTS). Configure Groq API key and test Llama 3.3 70B response. Register WhatsApp Business API — Meta verification takes 30 minutes. Deploy Polygon testnet wallet and fund with test MATIC. Set up eCourts NJDG API credentials. Set up Firebase project for push notifications.
3 PM – 5 PM	Architecture Scaffold	FastAPI project structure with all planned endpoints (stubs). ChromaDB connection tested end-to-end. React project with Tailwind, React Router, Zustand initialized. Capacitor initialized and Android build environment verified. All API keys stored in environment variables. GitHub repository with clean initial commit.
5 PM – 10 PM	Team Rest	Sleep. Eat properly. The 36-hour hackathon demands physical and mental endurance. A rested team at 6 PM on April 3rd outperforms an exhausted one every time.

8.2 Hours 0–12 — April 3, 6 PM to April 4, 6 AM

Period	Phase	Tasks and deliverables
Hours 0–4	RAG Voice Pipeline	Full voice loop: Sarvam STT → user query → ChromaDB retrieval → Groq LLM with retrieved law context → Hindi response → Sarvam TTS spoken back. Test with 10 different legal scenarios across wage, property, consumer, and criminal domains. Ensure citations are from actual statutes retrieved.

Hours 4–8	Document Generator + Blockchain	jsPDF generation for top 5 most common documents: unpaid salary notice, FIR draft, RTI application, illegal eviction reply, consumer forum complaint. WhatsApp share button. Polygon testnet hash write + transaction ID display in document footer. Verify end-to-end document flow.
Hours 8–12	Evidence Reader + Home UI	Multimodal image upload → Llama Vision API → structured JSON extraction → auto-fill legal notice form. Home dashboard screen, voice counsellor screen, document screen — all navigable with working bottom navigation. Amendment alert card on home screen with sample data.

8.3 Hours 12–24 — April 4, 6 AM to 6 PM (Mid-Eval at 3 PM)

Period	Phase	Tasks and deliverables
Hours 12–16	Document Decoder + Amendment Tracker	Clause extraction and risk classification pipeline. Green/amber/red card rendering. Counter-clause generation for red cards. Gazette scraper running (can use cached data for demo). Amendment diff view. User-specific amendment alert based on active case type.
Hours 16–20	Mid-Evaluation Preparation	Full rehearsal of Raju demo script. Fix all bugs identified in rehearsal. Polish voice response latency (target under 2 seconds). Ensure PDF looks professionally formatted. Ensure blockchain transaction ID displays correctly. Each team member knows exactly what to say and what to tap.
Hours 20–24	Case Tracker + NyayaScore	Case timeline UI with colored dots. Limitation period countdown for active case types. DLSA map with nearest office marker. NyayaScore algorithm implemented and gauge UI rendered. Score components showing individual document analysis. Fix any remaining bugs from mid-eval feedback.

8.4 Hours 24–36 — April 4, 6 PM to April 5, 6 AM

Period	Phase	Tasks and deliverables
Hours 24–28	WhatsApp Bot + ML Predictor	WhatsApp Business API webhook live and receiving messages. Message routing to FastAPI RAG backend. Response formatted for WhatsApp (no markdown, short paragraphs, reply buttons). Win probability ML predictor integrated into voice flow and displayed after case identification.
Hours 28–32	Negotiation Coach + Health Checkup Demo	Role-play conversation loop with employer/landlord persona injection. Post-session debrief LLM call. Legal Health Checkup — implement the monthly report generator (show demo output even if scheduler is not live). NyayaScore update trigger after document fixes.

Hours 32–36	Polish + Demo Rehearsal	UI polish across all screens (consistent spacing, no broken layouts). Fix all edge cases found in testing. Full 4-minute demo rehearsal 5 times. Assign speaker roles. Prepare 5 judge Q&A answers. Ensure GitHub commit history is clean and shows genuine development progression (judges check this). Submit on Devfolio before 5 AM deadline.
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8.5 Team Task Assignments

Team Member	Responsibilities
Shreeyash Kamble (Frontend + Voice)	React UI — all screens. Sarvam AI voice integration. jsPDF document generation. Android APK Capacitor build. Service worker offline setup. NyayaScore gauge UI. Multimodal evidence reader frontend.
Jatin Chenna (Backend + AI)	FastAPI backend architecture. ChromaDB RAG pipeline. Groq LLM integration. Blockchain Polygon integration. WhatsApp webhook. Win probability ML model. Gazette scraper. Overall system integration and GitHub management.
Ninad Shitole (Data + Integration + Demo)	Statute loading into ChromaDB. IPC-to-BNS mapping table. Indian Kanoon dataset collection and cleaning. State-specific law variants. Document template library (47 types). End-to-end testing. Demo script rehearsal. Devfolio submission.

CRITICAL RULE: Demo quality beats feature count every time.

A flawless 5-feature demo beats a broken 35-feature demo. If time runs short in hours 28-36, STOP adding features.

Spend every remaining hour making the Raju demo absolutely perfect.

One story, told perfectly, wins the hackathon.

SECTION 9 — ADDITIONAL INNOVATIONS (Beyond Current Plan)

9. Additional Innovation Ideas — Future Roadmap

These features are beyond the 36-hour hackathon scope but demonstrate the full vision of NyayaMitra as a product. Mentioning these during the presentation shows judges you are thinking beyond a hackathon project to a real societal solution.

[FUTURE — HIGH IMPACT] NyayaMitra for Police Stations — FIR Transparency

A secondary interface designed for police stations: any citizen can walk in, speak their complaint in their language, and the system generates a correctly formatted FIR draft using BNS sections — removing the police officer's ability to refuse registration due to 'not knowing which section applies.' This directly addresses one of India's most documented legal failures — police refusing to register FIRs. Can be implemented as a government-facing kiosk version of NyayaMitra.

Requires: Government partnership MoU | Separate police-facing UI | Section 173 BNSS compliance | Integration with CCTNS police records system

[FUTURE — TECHNICAL] Federated Learning for Privacy-Preserving Model Improvement

As millions of users use NyayaMitra, their anonymised legal pattern data becomes enormously valuable for improving the case outcome predictor and document quality. Federated learning allows model training on user data without the data ever leaving their device. Model updates are computed locally and only gradient updates (not data) are sent to the central server. This allows continuous model improvement while maintaining zero-knowledge privacy architecture.

Requires: TensorFlow Federated | Client-side model | Differential privacy | DPDP compliance framework

[FUTURE — ACCESSIBILITY] NyayaMitra IVR — Feature Phone Access via Phone Call

India has 300 million feature phone users who cannot use smartphone apps. An IVR (Interactive Voice Response) system allows them to call a toll-free number and access NyayaMitra's entire legal counselling pipeline through DTMF keypad and voice interaction. The call transcribes the user's spoken complaint, processes it through the RAG pipeline, and speaks back the legal guidance. Legal notices can be sent via SMS link.

Requires: Exotel / Twilio IVR | Voice activity detection | DTMF menu system | SMS document links | Toll-free number registration

[FUTURE — NETWORK EFFECT] Community Legal Knowledge Graph

Anonymised, aggregated patterns from millions of NyayaMitra cases build a legal knowledge graph specific to Indian conditions: which employers in which cities most frequently violate which labour laws, which landlord types most often attempt which illegal evictions, which consumer categories generate the most disputes. This data — shared with NGOs, journalists, and policymakers with user consent — becomes a real-time map of where legal rights violations are most concentrated in India. Transforms NyayaMitra from individual tool to systemic change infrastructure.

Requires: Anonymisation pipeline | Consent management | Graph database (Neo4j) | Public API for researchers | NGO partnership framework

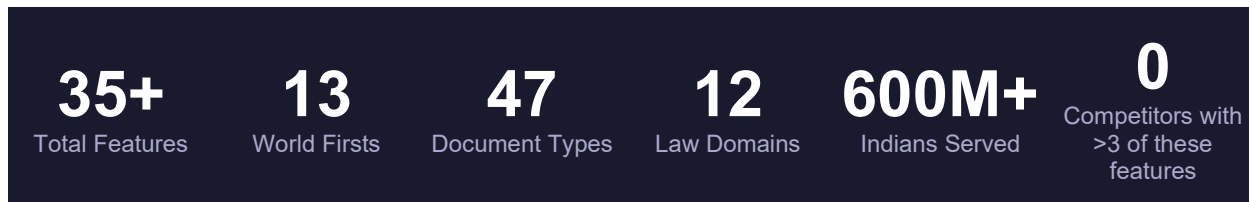
[FUTURE — MONETISATION] NyayaMitra Pro — Advocate Marketplace (Freemium)

For cases that exceed NyayaMitra's self-help capacity — complex high-court matters, criminal trials, property disputes above Rs. 50 lakh — users can upgrade to NyayaMitra Pro which connects them to a verified advocate network. Unlike LegalKart which is pure marketplace, NyayaMitra Pro users come with a complete pre-built case file: all facts extracted, relevant law sections identified, documents already drafted, case strategy outlined. Advocates on the platform can take the case knowing 80% of the groundwork is done. Lower advocate fees, better client preparation, better outcomes.

Requires: Advocate verification system | Bar Council API integration | Escrow payment system | Case handoff protocol | Revenue share model

SECTION 10 — FINAL SUMMARY

10. Why NyayaMitra Wins Hacksagon 2026



10.1 The Unbeatable Combination

- The problem is undeniable — 600 million Indians with no legal access is a humanitarian crisis, not a market opportunity
- The solution is complete — not a chatbot, not a template tool, not a lawyer directory — a full-stack AI legal justice platform
- The technology is real — self-built RAG, self-built ML, multimodal evidence reading, blockchain timestamping, Sarvam AI voice — this is not API-calling
- The world firsts are defensible — amendment live tracker, NyayaScore, AI negotiation coach, emotion-aware escalation, blockchain timestamping in legal tech — no competitor has these
- The demo is emotional — Raju's story is the story of 600 million Indians — judges will feel it, not just evaluate it
- The prototype works — live on Vercel, Android APK buildable, all core journeys demonstrable end-to-end

10.2 What NyayaMitra Is Not

- Not a chatbot with a legal theme — a full-stack AI system with 6 distinct ML/AI components
- Not an API wrapper project — self-built RAG, self-built ML outcome predictor, self-built gazette scraper
- Not English-only — 6 Indian languages, voice-first, designed for the semi-literate workforce
- Not reactive-only — proactive monthly legal health checkup, amendment alerts, limitation period warnings
- Not a hackathon toy — a product with a clear path to serving 600 million people

NyayaMitra does not give legal advice. It gives legal ACTION.

In your language. For free. In 60 seconds. To every Indian who needs it.

Team Return 0; — Hacksagon 2026 — ABV-IIITM Gwalior
Shreeyash Kamble | Jatin Chenna | Ninad Shitole